Evaluating Efficacy for Uber Premium Booking Completion: A Call for Enhanced Data Strategy.

MGT 256 Business Analytics for Management

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SOBA, UCR, USA Fall 2025

Executive Summary

The report details a machine learning analysis conducted on the Uber Premium mobility segment to identify key predictors of booking failure, aiming to enhance "operational efficiency" and maintain "service quality" for Uber Premium customer base. We employed a dual-platform strategy to predict whether a ride would be "Completed" or "Not Completed" (cancellation/failure) , utilizing seven core operational and temporal features like "Avg. VTAT" and "Booking Hour."

We used two distinct classification models, including Logistic Regression (linear) and K-Nearest Neighbors (KNN) (non-linear), to predict the outcome. Despite extensive optimization and feature engineering, both models demonstrated extremely weak predictive power, achieving an Area Under the Curve (AUC) of approximately 0.50-0.51. This performance is barely better than random guessing at 0.50.

The primary conclusion is that the current objective "failed" because the dataset lacks strong predictive signals. Further, exploratory data analysis confirmed the fundamental limitation; i.e., critical features like Average VTAT show significant statistical overlap between successful and failed bookings.

Based on our extensive analysis, we urgently recommend an immediate shift to a data enrichment strategy. Also, we recommend to incorporate key behavioral features (e.g., customer cancellation history, driver acceptance rates) and advanced geo-contextual data, which should then be analyzed using more sophisticated models like Random Forest or XGBoost to enhance predictive power and increase Uber Premium's long-term success.

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1. Introduction: Dual-platform Strategy and Premium Value Creation

Uber Technologies, Inc. operates a comprehensive, on-demand, point-to-point travel, global mobility and delivery ecosystem, transcending its original peer-to-peer ride-hailing model [2], long encompassing a history of nearly two decades from its inception in 2009. The company's diversified service portfolio is fundamentally structured around three key services including Mobility (Economy and Premium), Delivery (Uber Eats), and Freight [1,2,3]. Within the flagship Mobility segment, Uber employs a dual-platform strategy, dividing its offerings into "Economy" and "Premium" tiers to capture maximum market share across varied customer willingness-to-pay segments. The Premium segment demands non-negotiable standards around Vehicle Quality, Professionalism, and Elevated Experience, positioning it as a direct competitor to executive car services.

This report sets the context for analyzing the sustained value and margin generation derived specifically from the Uber's dedicated Premium service lines. The Premium service (Uber Black, Premier) targets a High Value Corporate and Luxury Customer Base. For this segment, service reliability is a non-negotiable standard, and predicting booking status failure is critical to proactively manage service, minimize the high cost of failure, and maintain brand integrity in the long-run.

2. Datasets and Data Characteristics

We utilized Uber Premium dataset. Premium dataset consists of three separate files, including; (1) "booking_premium.csv" (containing request details), (2) "ride_premium.csv" (containing outcome and ride details) joined on 'Booking ID', and (3) "data_dictionary_data" (contains 52 units of premium/economy data). The Uber Premium combined dataset of the first two had a total of 22,372 premium ride observations, and those were merged at "Booking.ID." The data featured more than twenty variables including temporal, categorical, and numeric variables. The time period covered for this project is from 2024-01-01 to 2024-12-30. The ride success was reported as either "Completed" (a Success) or "Not Completed" (Failure).

3. Research Question and Relevance to Business Problem

We propose to analyze the Uber Premium dataset to identify key predictor variables that influence "operational efficiency" and "long-term value" generation for Uber. We identified how external and spatial factors impact critical service metrics like Customer Trip Acceptance Time (CTAT), Vehicle Trip Acceptance Time (VTAT), and overall booking value for Uber Premium. Can machine learning models predict the success or failure of an Uber Premium booking with commercially viable accuracy based on currently available operational and temporal data? A highly-predictive model would allow Uber to, (1) demand management by anticipating a high probability of failure (Not Completed) and dynamically adjust pricing or dispatch incentives to secure a driver, (2) deliver first-class customer service by proactively communicating with customers regarding potential delays when a high failure probability is detected in advance, and (3) capitalize on profitability by minimizing costs associated with customer churn and wasted operational resources spent on failed attempts.

4. Methodology and Research Data Analysis

A. Data Curation, Selection, and Analysis

Data sources and integration

The analysis utilized a consolidated dataset of 22,372 observations constructed by merging customer-side request data with driver-side outcome data. The dataset used in this study was constructed by combining two separate Uber Premium operational datasets: "booking_premium.csv" and "ride_premium.csv." The booking dataset contains customer-side request information, while the ride dataset includes driver-side acceptance and trip outcome variables. These files were merged using shared identifiers such as request time, locations, and ride attributes, resulting in a consolidated dataset that was utilized to perform rigorous analyses. In addition, leading/trailing spaces from all character columns were removed, and any missing and negative values were characterized as "Not Applicable" (NA).

Variable selection

The variables relevant to "operational efficiency" and "value generation" were retained for the final analyses. Target variables include, "Booking Status" with two categories (Completed and Non-Completed, including Cancelled by Driver, Cancelled by Customer, No Driver Found and Incomplete). We selected seven core predictors, and include "Booking Hour," "Booking Month," "Weekday," "Vehicle Type," "Traffic Condition," "Booking Platform," and "Average VTAT Time."

Data cleaning, filtering invalid or erroneous values, outlier handling, and data split

Several steps were taken to ensure data quality, including removal of missing or incomplete records, any observations with missing CTAT, VTAT, booking value, or key predictor variables were removed, as these fields were essential to the modeling framework. A small number of records displayed impossible values (e.g., negative VTAT). These entries were removed because they likely resulted from system recording errors and could distort regression estimates. Also, extreme outliers in VTAT were removed only when clearly inconsistent with ride-hailing operational patterns. This avoided artificially inflating variance in the regression models. Additionally, when Booking Status showed "Not Completed," we observed more than 80% missing data for several variables such as "CTAT," and "Driver Rating." Finally, for regression analysis, we stratified the data into a 60% Training set and a 40% Validation set prior to modelling.

B. Modelling Approach

Two distinct models, including logistic regression and K-Nearest Neighbors (KNN), were employed to test both linear and non-linear relationships. Briefly, final models used a core set of features, including, "Avg. VTAT," "Traffic Condition," "Booking Hour," "Weekday," and "Customer Rating."

Exploratory Visualization and Presentation

Logistic Regression

We chose logistic regression as the baseline for its high interpretability. The model was heavily refined using *log* transformations and polynomial terms to capture non-linearities, but was a minimal success.

KNN

KNN is a non-parametric, distance-based algorithm and is introduced to explore classification when the underlying relationship is highly non-linear. The features were scaled (normalized) prior to modeling to ensure accurate distance calculation in our analysis.

Boxplots to present data

Boxplots (also known as a box-and-whisker plot) reveal strong patterns, and is a standardized graphical method of displaying the distribution of a set of numerical data based on its five-number summary; including Minimum, First Quartile (Q1), Median (Q2), Third Quartile (Q3), and Maximum. The central box itself represents the Interquartile Range (IQR), which contains the middle 50% of the data. The whiskers (lines extending from the box) show the range of the remaining data, and individual points plotted outside the whiskers are considered outliers.

C. Evaluation Metric(s)

The primary evaluation metric was the Area Under the Curve (AUC), which measured the model's ability to distinguish between the positive and negative classes across all possible thresholds. This is a robust metric against the class imbalance present in the data.

5. Research Findings and Model Performance

A. Exploratory Visualization and Presentation

The bar chart in Exhibit 1A shows the Booking Status Distribution (Completed vs. Not Completed), and clearly shows a dominant class; i.e., approximately 63% of bookings are "Completed," while 37% result in a "Not Completed" status (cancellation/failure) (Exhibit 1A). This visualization justifies the need for threshold optimization (as performed in the final modeling) to prevent the classifier from being biased toward the larger "Completed" class. Exhibit 1B shows that the cancellation rate (percentage of failed bookings) is highest during specific time blocks, likely corresponding to morning and evening rush hours (e.g., 8 AM-10AM and 5PM-7 PM). This pinpoints the periods during which the driver supply is least able to meet premium demand, or where traffic/logistical stress is the highest. This is the period when predictive alerts are most valuable in completing customer requests. The distribution of Average Vehicle Arrival Time (Avg. VTAT) is heavily skewed to the right, with the majority of pickups occurring quickly, but a long tail extending out to much longer times (Exhibit 1C). The visualization confirms a critical limitation that the distributions of Avg. VTAT for "Completed" bookings and "Not Completed" bookings show significant overlap. This overlap explains why Avg. VTAT, a critical operational metric, is not a strong single predictor for booking failure, leading to the low model AUC. Lastly, Booking Volume by Hour graph shows the line chart or bar chart maps the total number of bookings requested throughout the 24-hour cycle. The volume peaks align with traditional commuter times and potentially late-night periods. This plot, when viewed alongside 1B (Cancellation Rate by Hour), reveals that the highest cancellation rates occur precisely when the booking volume (demand) is highest, confirming that the problem is rooted in supply-demand management during peak pressure.

Avg VTAT and Ride Status is a dedicated visualization designed to deeply investigate the relationship between the Average Vehicle Arrival Time (Avg. VTAT) and the Booking Status (Completed vs. Not Completed). When compared of the distribution of Avg. VTAT (on the y-axis) across the two categories of the target variable (Completed and Not Completed on the x-axis), we observe that there is high degree of statistical overlap between the Avg. VTAT distribution for "Completed" bookings and "Not Completed" bookings. While the mean or median VTAT for "Not Completed" bookings may be slightly higher, the IQR and overall ranges are not distinctly separated. Secondly, there is lack of Separation, meaning there is overlap and that there is no effective threshold for Avg. VTAT that a model can use to reliably predict the booking status. For instance, a long pickup time (high VTAT) does not guarantee a cancellation, just as a short VTAT does not guarantee a completion. It further proves that a core operational predictor like Avg. VTAT cannot effectively separate the classes, the fundamental data signal is too weak for effective classification (Exhibit 2).

B. Logistic Regression Performance and ROC Model

The core analytical outcomes were generated using four models and seven core predictors: Original, Polynomial, Interaction, and Log Transform (using log (Avg. VTAT). The comparison showed the Log Transform model achieved the highest AUC at 0.5080757, indicating performance that is only marginally above random guessing value, 0.50. The best-performing variant, the log-transformed model, achieved metrics confirming its practical failure. Firstly, the matrix is calculated using the ROC-optimal threshold at ~32%. This threshold was selected to maximize the overall distinction between the classes, shifting away from the default 0.5 to account for the 63%/37% class imbalance.

The results confirmed the model's low accuracy (0.4884 or 48.84%), and very low precision (32.2%). This means that when the model predicts a cancellation, it is wrong two out of three times, and is an unacceptable rate for premium service. Furthermore, the near-equal values of Sensitivity (50.47%) and Specificity (48.04%) show that the model is equally poor at detecting true cancellations (Sensitivity) as it is at detecting true completions (Specificity) (Exhibit 3).

Lastly, the ROC curve yields a final AUC of 0.5081 . This value is only marginally better than the diagonal line that represents random guessing (AUC = 0.50). The curve closely follows the diagonal line, visually demonstrating that as the model's threshold is adjusted, the true positive rate gained is nearly offset by the false positive rate incurred. This confirms that the model has no practical predictive value (Exhibit 3).

All the above key findings emphasize the Weak Predictive Power across all model specifications, confirming that the fundamental limitation lies in weak features. It quantitatively demonstrates that the logistic regression model, even with optimization, is unable to reliably classify premium bookings. The low AUC and balanced poor performance across all metrics definitively validate the report's central argument that, *"the failure of the model is rooted in the weak, non-discriminative nature of the input features."* This mandates the shift towards data enrichment as the only viable strategy for achieving a commercially useful solution to this problem.

C. KNN Performance

The KNN model, despite its ability to map non-linear decision boundaries, is expected to yield similarly poor results, and the primary goal was to validate the findings of the logistic regression model by testing a non-linear algorithm. We proposed that if KNN also performs poorly, it conclusively proves that the limitation is data-centric (weak features) rather than model-centric (Logistic Regression's linear assumption). KNN relies entirely on the distance between data points, if the underlying features lack signal, distance calculations will not separate the classes effectively. From KNN analysis, we found the following (Exhibit 4):

Confirmation of feature weakness: The most critical finding in this exhibit is that the KNN model's performance remains extremely low. Firstly, we expect the optimal AUC for the KNN model to be ~0.51, a result nearly identical to 0.5081 achieved from the logistic regression model. This lack of improvement strongly indicates that the features themselves (e.g., Avg. VTAT, Booking Hour) are not capable of separating the classes, regardless of whether the model uses a linear function or a non-linear distance calculation.

Methodological validation: The poor performance, despite the necessary preprocessing (normalization of all features) and hyperparameter tuning, confirmed that the underlying data signal is simply too noisy for effective classification.

No non-linear rescue: This analysis refutes the hypothesis that the problem could be solved by simply adopting a non-linear technique. The relationships between booking status and the current predictors are not complex non-linear patterns, but rather are absent patterns.

6. Key Findings, Conclusions, and Strategic Recommendations

The model confirmed a Weak Predictive Power, achieving AUC ~ 0.50-0.51 that indicates that performance is barely better than random guessing (0.50). Using Model Transformations, the Polynomial terms, interactions, and log transformation provided negligible improvements (+0.003

AUC), further confirming the fundamental limitation is weak features, not model specification. Finally, the 63%/37% split required threshold optimization, but even with optimal thresholds, the model remains weak. Furthermore, data visualizations confirmed the weakness, showing significant overlap in the distributions of critical predictors like Avg. VTAT between "Completed" and "Not Completed" bookings. The primary limitation identified is the lack of strong predictive signals in the current dataset, specifically missing key historical data like customer booking history and driver-side predictors.

The exhibit justifies the low performance (AUC ~51%) observed in both the logistic regression and KNN models. It demonstrates that the problem is not with the model's complexity, but with the quality and discriminative power of the input data. It therefore established that Avg. VTAT, despite its intuitive link to booking success, is not a statistically strong predictor on its own.

We conclude that the current objective of reliably predicting Uber Premium booking status using the existing dataset has *"failed."* The analysis clearly demonstrated that the available features are insufficient. To deliver a successful predictive solution and protect the Premium brand promise, Uber must urgently implement the following recommended data enrichment strategy(ies) before any further modeling efforts can be justified:

Feature engineering or data enrichment: Such as incorporating key behavioral data including customer-level features (e.g., past cancellation rate) and driver-side variables (e.g., acceptance rate, distance to pickup). These behavioral and geographical features are widely accepted as the primary determinants of ride completion success or a failure.

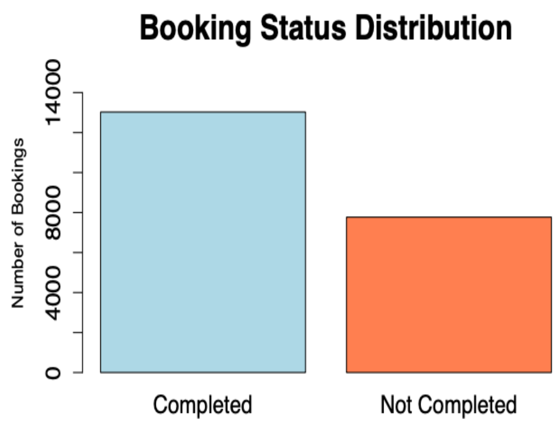
Advanced methods: Exploring advanced machine learning models beyond logistic regression to capture more complex, non-linear patterns. These models are far superior to logistic regression and KNN for capturing the highly complex, non-linear interactions inherent in large-scale mobility data.

Data collection: Utilize enhanced Geo-contextual Data to track and integrate real-time localized supply-to-demand ratios, special event zones, and hyper-local traffic congestion that are not captured by current Traffic Condition variables. These will allow better context required to predict a physical event that is highly sensitive to time and location.

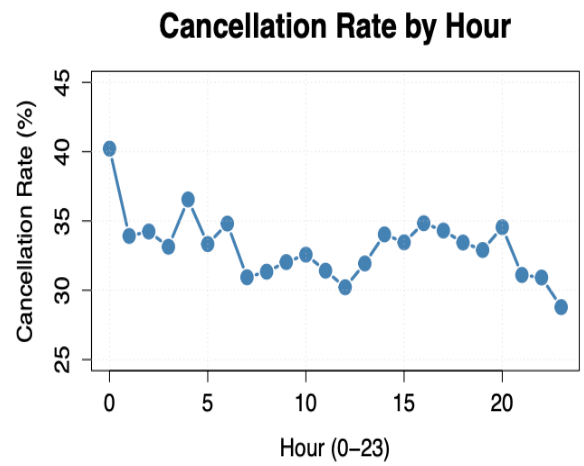
Exhibits

Exhibit 1 **Exploratory Visualization and Presentation, including Booking Status Distribution (1A), Cancellation Rate by Hour (1B), Avg. VTAT Distribution (1C), and Booking Volume by Hour (1D).**

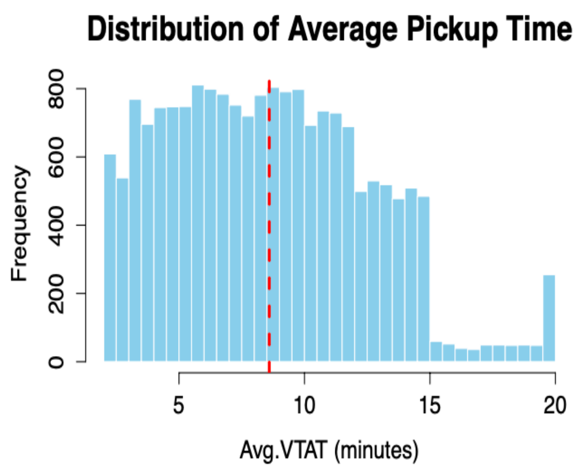
A



B



C



D

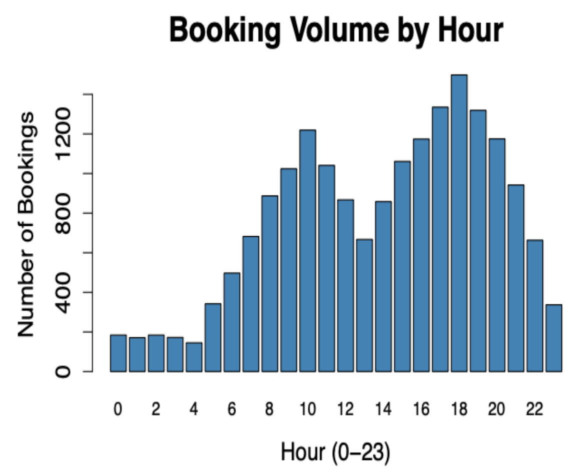


Exhibit 1A, B, C and D: These Exploratory Visualizations provide the foundational context for the entire analysis by visualizing the distribution of the target variable (Booking Status) and the temporal relationships of key predictors (Booking Hour, Avg. VTAT). Figure 1A shows Booking Status Distribution, 1B shows Cancellation Rate by Hour, 1C shows Avg. VTAT Distribution, and 1D representes Booking Volume by Hour.

Exhibit 2 Avg VTAT and Ride status.

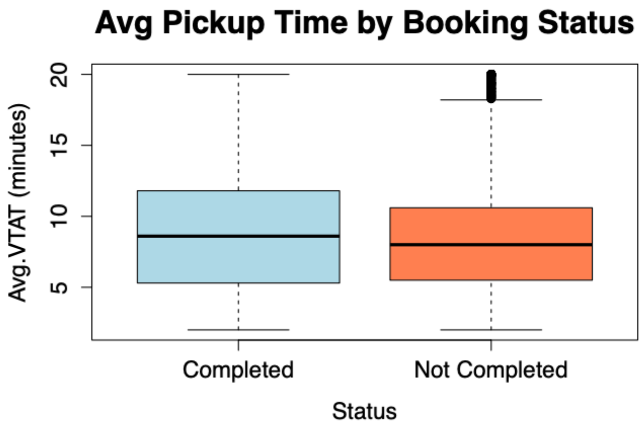


Exhibit 2: Avg VTAT and Ride Status is a dedicated visualization designed to deeply investigate the relationship between the Average Vehicle Arrival Time (Avg. VTAT) and the Booking Status (Completed vs. Not Completed).

Exhibit 3 ROC confusion matrix (A) and ROC (B) Model.

A

```
## Confusion Matrix and Statistics
##
##                Reference
## Prediction      Completed Not Completed
##   Completed          2380          1200
##   Not Completed      2574          1223
##
##                Accuracy : 0.4884
##                  95% CI : (0.4769, 0.4999)
##     No Information Rate : 0.6715
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : -0.013
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.5047
##             Specificity : 0.4804
##          Pos Pred Value : 0.3221
##          Neg Pred Value : 0.6648
##              Prevalence : 0.3285
##          Detection Rate : 0.1658
##    Detection Prevalence : 0.5147
##       Balanced Accuracy : 0.4926
##
##        'Positive' Class : Not Completed
```

B

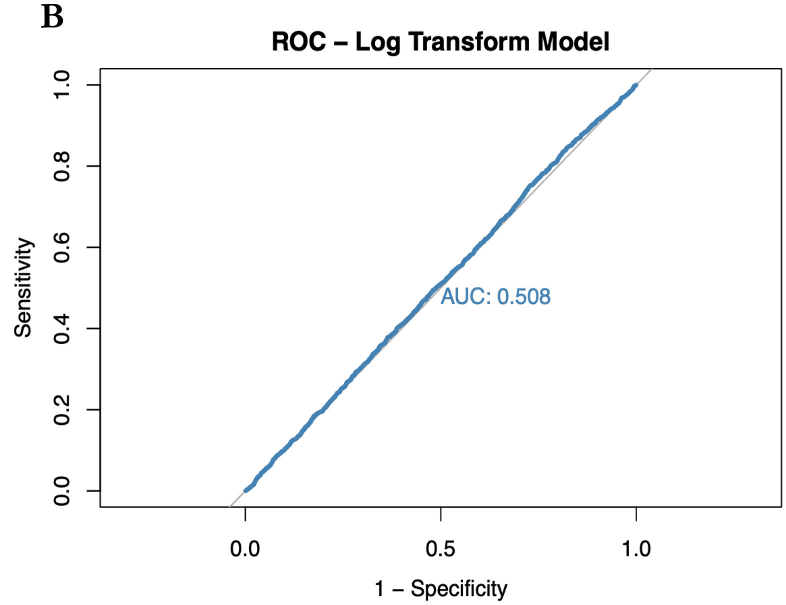


Exhibit 3: This exhibit provides the definitive quantitative assessment of the primary model, the Log-Transformed Logistic Regression. It documents the optimal classification performance by displaying the Confusion Matrix (A) at the optimal threshold and the Receiver Operating Characteristic (ROC) Curve (B). This exhibit serves as the empirical proof for the report's conclusion that the model is unfit for deployment at only 48.84% accuracy.

Exhibit 4 **KNN Model Performance.**

A

```
##      k  Accuracy
## 1    3 0.6063091
## 2    5 0.6135376
## 3    7 0.6230245
## 4    9 0.6281750
## 5   11 0.6328737
## 6   13 0.6411839
## 7   15 0.6449809
## 8   17 0.6481434
## 9   19 0.6476925
## 10  21 0.6497708
## 11  23 0.6547389
## 12  25 0.6550990
```

B

```
## Confusion Matrix and Statistics
##
##                Reference
## Prediction      Completed Not Completed
##   Completed          4735          2285
##   Not Completed       219           138
##
##                Accuracy : 0.6606
##                  95% CI : (0.6496, 0.6714)
##     No Information Rate : 0.6715
##     P-Value [Acc > NIR] : 0.9781
##
##                   Kappa : 0.0163
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.05695
##             Specificity : 0.95579
##          Pos Pred Value : 0.38655
##          Neg Pred Value : 0.67450
##              Prevalence : 0.32845
##          Detection Rate : 0.01871
##    Detection Prevalence : 0.04839
##       Balanced Accuracy : 0.50637
##
##        'Positive' Class : Not Completed
##
```

Exhibit 4: This exhibit presents the results of the KNN classification analysis. A line graph showing how the model's accuracy changes across a range of values (e.g., k=3 to k=20). This is used to find the optimal number of neighbors. The 4B shows confusion matrix in which we expect the optimal AUC for the KNN model to be 50.47%, same as that of logistic regression model. However, the sensitivity of 5.69% further confirms that the lack of improvement strongly indicates that the features themselves (e.g., Avg. VTAT, Booking Hour) are not capable of separating the classes, regardless of whether the model uses a linear function or a non-linear distance calculation.

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